compact yet detailed cheat sheet of the **most common Java Collection methods** with descriptions, grouped by collection type. This is perfect for interviews or quick reference.

1️⃣ ArrayList / List

- **add(E e)** → Adds an element to the end of the list.
- **get(int index)** → Returns the element at the given index.
- **set(int index, E element)** → Replaces the element at the given index.
- **remove(int index) / remove(Object o)** → Removes element by index or value.
- **size()** → Returns the number of elements.
- **contains(Object o)** → Returns true if the list contains the element.
- **isEmpty()** → Checks if the list is empty.
- **clear()** → Removes all elements from the list.

2️⃣ LinkedList / Deque

- **addFirst(E e) / addLast(E e)** → Adds element to start or end.
- **removeFirst() / removeLast()** → Removes element from start or end.
- **getFirst() / getLast()** → Returns first or last element.
- **offer(E e)** → Adds element to the queue (like add).
- **poll()** → Retrieves and removes the head of the queue.
- **peek()** → Retrieves but does not remove the head.
- **size()**, **isEmpty()**, **clear()** → Standard list/collection methods.

3️⃣ HashSet / Set

- **add(E e)** → Adds element (ignored if duplicate).
- **remove(Object o)** → Removes the element.
- **contains(Object o)** → Checks if element exists.
- **size()** → Number of elements.

- **isEmpty()** → True if empty.
- **clear()** → Removes all elements.

LinkedHashSet preserves **insertion order**, TreeSet stores elements in **sorted order**.

4️⃣ HashMap / Map

- **put(K key, V value)** → Inserts or updates key-value pair.
- **get(K key)** → Returns value for the key (null if not found).
- **remove(K key)** → Removes key-value pair.
- **containsKey(K key)** → Checks if a key exists.
- **containsValue(V value)** → Checks if a value exists.
- **keySet()** → Returns a set of all keys.
- **values()** → Returns a collection of all values.
- **entrySet()** → Returns a set of Map.Entry (key-value pairs).
- **size()**, **isEmpty()**, **clear()** → Standard operations.

LinkedHashMap preserves **insertion order**, TreeMap maintains **sorted keys**.
TreeMap also has: **firstKey()**, **lastKey()**, **ceilingKey(K key)**, **floorKey(K key)**.

5️⃣ PriorityQueue

- **offer(E e) / add(E e)** → Adds element to the queue.
- **poll()** → Retrieves and removes the head (min or max depending on comparator).
- **peek()** → Retrieves but does not remove the head.
- **remove(Object o)** → Removes specific element.
- **size()**, **isEmpty()**, **clear()** → Standard methods.

Min-heap by default; for max-heap: new PriorityQueue<>(Collections.reverseOrder()).

🔹 Easy Memory Tip / Mnemonic

Collection	Mnemonic	Methods
ArrayList	**AGSR**	Add, Get, Set, Remove, Size, contains, isEmpty, clear
LinkedList / Deque	**FL-OPP**	addFirst/Last, removeFirst/Last, offer, poll, peek
Set	**ARCS**	Add, Remove, Contains, Size, isEmpty, clear
Map	**PGR-CKVE**	Put, Get, Remove, Contains (Key/Value), keySet, values, entrySet
Tree	**Sorted**	first(), last(), ceiling(), floor()
PriorityQueue	**OPP**	Offer, Poll, Peek, remove, size, isEmpty, clear

2 Strings

Method	Description
charAt(i)	Returns character at index i
length()	Returns string length
substring(start, end)	Returns substring from start to end-1
contains(CharSequence)	Checks if string contains sequence
indexOf(char/str)	Returns first index of char or substring
lastIndexOf(char/str)	Returns last index
startsWith(str) / endsWith(str)	Checks prefix or suffix
equals(str) / equalsIgnoreCase(str)	Compares strings
replace(old, new)	Replaces characters or substring
split(regex)	Splits string into array
toCharArray()	Converts to char array
trim()	Removes leading/trailing spaces
toUpperCase() / toLowerCase()	Converts case
concat(str)	Concatenates strings

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3 Arrays

Method	Description
Arrays.sort(arr)	Sorts array
Arrays.copyOf(arr, len)	Copies array with length
Arrays.copyOfRange(arr, from, to)	Copies range
Arrays.fill(arr, value)	Fills array with value
Arrays.equals(arr1, arr2)	Compares two arrays
Arrays.toString(arr)	Converts to string
Arrays.binarySearch(arr, key)	Binary search in sorted array

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4 Math Class

Method	Description
Math.max(a,b)	Maximum of two numbers
Math.min(a,b)	Minimum
Math.abs(x)	Absolute value
Math.sqrt(x)	Square root
Math.pow(x,y)	x raised to y
Math.ceil(x)	Smallest integer ≥ x
Math.floor(x)	Largest integer ≤ x
Math.round(x)	Rounds to nearest integer
Math.random()	Random double [0,1)

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5 Character Class

Method	Description
isDigit(c)	Checks if character is a digit

Method	Description
isLetter(c)	Checks if character is a letter
isUpperCase(c) / isLowerCase(c)	Case check
toUpperCase(c) / toLowerCase(c)	Convert case

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6 System & Utility

Method	Description
System.currentTimeMillis()	Current time in ms
System.out.println()	Print line
System.arraycopy(src, srcPos, dest, destPos, length)	Copy array
Collections.sort(list)	Sort a list
Collections.reverse(list)	Reverse a list
Collections.max(list) / Collections.min(list)	Max/min of list